

Desmos Mastery - Day 2

24 original questions | Calculator permitted

Complete the questions before turning to the answer explanations. Mark any guess, even if it turns out to be correct. For math, show your setup and check the requested quantity. For reading, identify the exact supporting detail; for grammar, identify the clause or subject.

These questions were authored for this workbook from the lesson topics. They are not official SAT questions or copies of the course's paid practice.

Single Variable Equations | 7 questions

[Lesson video: Single Variable Equations](#)

1. What is the greater solution of the equation below?

$$x^2 - 5x - 24 = 0$$

Answer: _____

2. If x is the positive solution of the equation below, what is $x/4$?

$$|x - 3| = 9$$

Answer: _____

3. The equation below has one solution in its domain. What is it?

$$\frac{3x+1}{x-2} = 4$$

Answer: _____

4. Which value solves the equation below?

$$\sqrt{x+5} = x-1$$

- A) -1
- B) 0
- C) 4
- D) 9

5. A quantity is modeled by the expression below. At what value of t does the quantity equal 54?

$$Q(t) = 6(3)^t$$

Answer: _____

6. The smaller root of $x^2 - 10x + k = 0$ is 4. What is k ?

Answer: _____

7. The equation below has a solution x . What is $x + 1$?

$$\frac{2}{x-2} = \frac{3}{x+1}$$

Answer: _____

Systems of Equations | 10 questions

[Lesson video: Systems of Equations](#)

8. What is the value of $x + y$ for the system below?

$$x + 2y = 11$$

$$3x - y = 12$$

Answer: _____

9. The three equations below have a common solution (x, y) . What is k ?

$$x + y = 7$$

$$x - y = 1$$

$$2x + 2y = k$$

Answer: _____

10. Which point is a solution of all three equations?

$$\begin{aligned}x + y &= 5 \\ 2x - y &= 1 \\ xy &= 6\end{aligned}$$

- A) (2, 3)
- B) (3, 2)
- C) (1, 4)
- D) (4, 1)

11. For every real solution of the system below, what is $x^2 + y^2$?

$$\begin{aligned}x + y &= 7 \\ xy &= 10\end{aligned}$$

Answer: _____

12. The line $y = 2x + 1$ intersects the parabola $y = x^2 - 2x + 1$ at two points. What is the larger x -coordinate?

Answer: _____

13. A system has the unique solution shown by the equations below. What is y ?

$$\begin{aligned}x &= 3y - 5 \\ 2x + y &= 18\end{aligned}$$

Answer: _____

14. What is the solution set of the three equations below?

$$\begin{aligned}x + y &= 5 \\x - y &= 1 \\3x + y &= 12\end{aligned}$$

- A) Only (3, 2)
- B) Only (2, 3)
- C) Infinitely many points
- D) No solution

15. If x and y satisfy the equations below, what is $2^x/4^y$?

$$\begin{aligned}x - 2y &= 3 \\x + y &= 12\end{aligned}$$

Answer: _____

16. The system below has two real solutions. What is the sum of their x -coordinates?

$$\begin{aligned}x + y &= 5 \\xy &= 4\end{aligned}$$

Answer: _____

17. The equations below have a common solution. What is c ?

$$\begin{aligned}3x - y &= 4 \\x + y &= 8 \\x + 2y &= c\end{aligned}$$

Answer: _____

Number of Solutions | 7 questions

[Lesson video: Number of Solutions](#)

18. How many real solutions does the equation below have?

$$x^2 - 6x + 9 = 0$$

- A) 0
- B) 1
- C) 2
- D) Infinitely many

19. How many real solutions does the equation below have?

$$(x + 2)^2 + 4 = 0$$

- A) 0
- B) 1
- C) 2
- D) Infinitely many

20. How many solutions does the system below have?

$$2x - 3y = 6$$

$$6x - 9y = 18$$

- A) 0
- B) 1
- C) 2
- D) Infinitely many

21. How many solutions does the system below have?

$$2x - 3y = 6$$

$$4x - 6y = 14$$

- A) 0
- B) 1
- C) 2
- D) Infinitely many

22. For what value of k does $y = x^2 - 4x + 7$ and $y = k$ have exactly one common point?

Answer: _____

23. How many real solutions does the equation below have?

$$|x - 2| = \frac{x}{2} + 1$$

- A) 0
- B) 1
- C) 2
- D) 3

24. How many real solutions does the equation below have?

$$3(x + 2) = 3x + 6$$

- A) 0
- B) 1
- C) 2
- D) Infinitely many

Answers & explanations

Count correct answers, but also review correct guesses. Redo missed questions after a day without looking at the solution.

1: 8 2: 3 3: 11 4: C 5: 2 6: 24 7: 9 8: 8 9: 14 10: A 11: 29 12: 4 13: 4 14: D 15: 8 16: 5 17: 13 18: B 19: A 20: D 21: A 22: 3 23: C 24: D

Why each answer works

- 1. Answer: 8.** The quadratic factors as $(x - 8)(x + 3) = 0$, giving -3 and 8 .
- 2. Answer: 3.** The solutions are $x = 12$ and $x = -6$. Use the positive solution and report $12/4 = 3$.
- 3. Answer: 11.** Multiply by $x - 2$, which cannot be zero: $3x + 1 = 4x - 10$. Thus $x = 11$, which is permitted.
- 4. Answer: C.** Squaring gives $x + 5 = (x - 1)^2$, so $x = -1$ or 4 . The original right side must be nonnegative; -1 is extraneous. At $x = 4$ both sides equal 3 .
- 5. Answer: 2.** Divide by 6 to obtain $3^t = 9$, so $t = 2$.
- 6. Answer: 24.** Substitute 4 : $16 - 40 + k = 0$, so $k = 24$. The other root is 6 , confirming that 4 is smaller.
- 7. Answer: 9.** Multiply by $(x - 2)(x + 1)$, excluding 2 and -1 : $2(x + 1) = 3(x - 2)$. Thus $x = 8$, and $x + 1 = 9$.
- 8. Answer: 8.** The first equation gives $x = 11 - 2y$. Substitution yields $3(11 - 2y) - y = 12$, so $y = 3$ and $x = 5$. Therefore $x + y = 8$.
- 9. Answer: 14.** The first two equations give $x = 4$ and $y = 3$. Substituting into the third gives $k = 2(4) + 2(3) = 14$.
- 10. Answer: A.** Every option has $x + y = 5$. Only $(2, 3)$ also has $2x - y = 1$ and $xy = 6$. Checking just one or two equations is insufficient.
- 11. Answer: 29.** Expand $(x + y)^2 = x^2 + 2xy + y^2$. Therefore $x^2 + y^2 = 49 - 20 = 29$. Either ordered solution gives the same value.
- 12. Answer: 4.** Set the expressions equal: $x^2 - 4x = 0$. The x -coordinates are 0 and 4 .
- 13. Answer: 4.** Substitution gives $2(3y - 5) + y = 18$, so $7y = 28$ and $y = 4$.
- 14. Answer: D.** The first two require $x = 3$ and $y = 2$. At that point $3x + y = 11$, not 12 , so the three equations have no common solution.
- 15. Answer: 8.** Rewrite 4^y as 2^{2y} . The ratio is $2^x - 2^y = 2^3 = 8$. You do not need the individual values.
- 16. Answer: 5.** Substitute $y = 5 - x$ into $xy = 4$ to get $x^2 - 5x + 4 = 0$. Its roots are 1 and 4 , whose sum is 5 .
- 17. Answer: 13.** The first two imply $y = 3x - 4$ and $y = -x + 8$. Thus $x = 3$, $y = 5$, and $c = x + 2y = 13$.
- 18. Answer: B.** The equation becomes $(x - 3)^2 = 0$. A repeated root is one distinct real solution.
- 19. Answer: A.** The left side is at least 4 for every real x , so it cannot equal zero.

20. Answer: D. Dividing the second equation by 3 gives the first. They represent the same line, so every point on that line solves both.

21. Answer: A. The second equation reduces to $2x - 3y = 7$, which contradicts $2x - 3y = 6$. These are parallel distinct lines.

22. Answer: 3. The parabola is $(x - 2)^2 + 3$. A horizontal line touches it once only at the vertex height, $k = 3$.

23. Answer: C. Split at $x = 2$. For $x \geq 2$, $x - 2 = x/2 + 1$ gives $x = 6$. For $x < 2$, $2 - x = x/2 + 1$ gives $x = 2/3$. Both satisfy their cases.

24. Answer: D. Each side expands to $3x + 6$. This is an identity, so every real x is a solution. A graph display alone may not distinguish an identity from an inconsistent equation.

For each missed or guessed question, record the skill, the condition you overlooked, and one specific action for next time.